

Stability and Convergence of one-step methods

$$u'(t) = f(u) \quad t \in [0, T]$$

$$u(0) = \eta$$

f : Lipschitz continuous on $D \in \eta$

$$N = T/k \quad U^n \approx u(t_n = nk)$$

$$E^N = U^n - u(t_n)$$

We want to show
that

$$\lim_{k \rightarrow 0} \|E^N\| = 0$$

(The limit is taken with T
fixed, so $N \rightarrow \infty$)

Proof in 3 steps:

- ① Euler's method applied to linear, scalar problem
- ② Euler's method applied to a nonlinear system
- ③ Runge-Kutta method applied to a nonlinear system

$$u'(t) = \lambda u(t) + g(t)$$

$$u(0) = \eta$$

$$\text{Solution: } u(t) = e^{\lambda t} \eta + \int_0^t e^{\lambda(t-\tau)} g(\tau) d\tau$$

(Duhamel's principle)

Euler's method:

$$U^{n+1} = U^n + k\lambda U^n + kg(t_n)$$

$$u(t_{n+1}) = u(t_n) + k\lambda u(t_n) + kg(t_n) + k\tau^n$$

$$E^{n+1} = E^n + k\lambda E^n - k\tau^n$$

$$\tau^n = O(k)$$

$$E^{n+1} = (1+k\lambda)E^n - k\tau^n$$

$$E^n = (1+k\lambda)E^{n-1} - k\tau^{n-1}$$

Substitute:

$$E^{n+1} = (1+k\lambda)^2 E^{n-1} - (1+k\lambda)k\tau^{n-1} - k\tau^n$$

...

$$E^N = (1+k\lambda)^N E^0 - k \sum_{m=1}^N (1+k\lambda)^{N-m} \tau^{m-1}$$

Analogous to Duhamel's principle

$$N = \frac{T}{k}$$

$$kN = T$$

$$|E^N| = \left| k \sum_{m=1}^N (1+k\lambda)^{N-m} \tau^{m-1} \right|$$

$$\leq k \sum_{m=1}^N |1+k\lambda|^{N-m} |\tau^{m-1}|$$

Lemma:

$$|1+k\lambda| \leq |1+k|\lambda| \leq \sum_{j=0}^{\infty} \frac{k^j |\lambda|^j}{j!} = e^{k|\lambda|}$$

$$|E^N| \leq k \sum_{m=1}^N e^{k|\lambda|(N-m)} |\tau^{m-1}|$$

$$\leq k \sum_{m=1}^N e^{kN|\lambda|} |\tau^{m-1}|$$

$$\leq Nk e^{T|\lambda|} \max_{1 \leq m \leq N} |\tau^{m-1}|$$

$$|E^N| \leq T e^{T|\lambda|} \|\tau\|_\infty$$

So $|E^N| \rightarrow 0$ as $k \rightarrow 0$.

Let $T=10$, $\lambda=-10$.

We have

$$|E^N| \leq 10e^{100} \|\tau\|_\infty$$

$\Rightarrow$ totally useless!

② Euler's method applied to a nonlinear system

$$u'(t) = f(u) \quad u: \mathbb{R} \rightarrow \mathbb{R}^m$$

$$u(0) = \eta$$

$$\|f(v) - f(w)\| \leq L \|v - w\| \quad \forall v, w \in D$$

We have:

$$U^{n+1} = U^n + k f(U^n)$$

$$u(t_{n+1}) = u(t_n) + k f(u(t_n)) + k \tau^n$$

$$E^{n+1} = E^n + k (f(U^n) - f(u(t_n))) - k \tau^n$$

$$\begin{aligned} \|E^{n+1}\| &\leq \|E^n\| + k\|f(U^n) - f(u(t_n))\| + k\|\tau^n\| \\ &\leq \|E^n\| + kL\|U^n - u(t_n)\| + k\|\tau^n\| \end{aligned}$$

$$\|E^{n+1}\| \leq (1+kL)\|E^n\| + k\|\tau^n\|$$

Substitute into itself

$$\|E^N\| \leq (1+kL)^N \|E^0\| + k \sum_{m=1}^N (1+kL)^{N-m} \|\tau^{m-1}\|$$

Follow the steps of the previous proof, with L in place of λ . We obtain

$$\|E^N\| \leq T e^{TL} \max_{0 \leq m < N+1} \|\tau^m\|$$

$$\text{So } \lim_{k \rightarrow 0} \|E^N\| = 0.$$

③ Convergence of RK methods

Example:

$$U^* = U^n + \frac{1}{2}k f(U^n)$$

$$U^{n+1} = U^n + k f(U^*)$$

Rewrite:

$$\frac{U^{n+1} - U^n}{k} = \underbrace{f\left(U^n + \frac{1}{2}k f(U^n)\right)}_{\Phi(U^n)}$$

Claim: If f is L.C., then

Ψ is also L.C.

Proof: Assume

$$\|f(v) - f(w)\| \leq L\|v - w\|$$

$$\text{Then: } \|\Psi(v) - \Psi(w)\| = \|f(v + \frac{1}{2}kf(v)) - f(w + \frac{1}{2}kf(w))\|$$

$$\leq L\|v + \frac{1}{2}kf(v) - w - \frac{1}{2}kf(w)\|$$

$$\leq L\|v - w\| + \frac{1}{2}kL^2\|f(v) - f(w)\|$$

$$\leq L\|v - w\| + \frac{1}{2}kL^2\|v - w\|$$

$$\leq (L + \frac{1}{2}kL^2)\|v - w\|$$

So Ψ is L.C. with constant $L + \frac{1}{2}kL^2$

So we have

$$U^{n+1} = U^n + k\Psi(U^n)$$

We follow the previous proof, with

$$f \rightarrow \Psi$$

$$L \rightarrow L + \frac{1}{2}kL^2$$

We get

$$\|E^N\| \leq T e^{T(L + \frac{1}{2}kL^2)} \max_m \|z^m\|$$

With a different RK
method, the only change
is the L.C. of Ψ .